Chapter 3. Entropy, KL Divergence and Cross-Entropy

Lectures: Week 1 Wed, Week 2 Mon–Wed (Entropy, KL, mutual information, cross entropy); PS1 Problem 1

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1 Information and Entropy

The information of an outcome is $h(x) = -\log p(x)$: rare outcomes carry more information, and for independent events probabilities multiply while information adds.

Definition: entropy

For a discrete random variable with pmf p,

$$H(X) = -\sum_x p(x)\log p(x) = \mathbb{E}[-\log p(X)], \qquad 0\log 0 := 0.$$

Natural logs give nats; base 2 gives bits.

Theorem: range of entropy

If X takes at most K values, $0 \le H(X) \le \log K$. $H = 0$ iff X is deterministic; $H = \log K$ iff X is uniform.

Proof. Each term $-p\log p \ge 0$ for $p \in [0,1]$, with equality iff $p \in \{0,1\}$. For the upper bound apply Jensen to the concave log: $H = \mathbb{E}\log\frac{1}{p(X)} \le \log \mathbb{E}\frac{1}{p(X)} = \log\sum_{x:p(x)>0} 1 \le \log K$, with equality iff $1/p(X)$ is constant on the support and the support has K points. ■

(For continuous densities the differential entropy can be negative; the statement above is for discrete variables.)

Definition: joint and conditional entropy

$$H(X,Y) = -\sum_{x,y} p(x,y)\log p(x,y),$$

$$H(Y\mid X) = -\sum_{x,y} p(x,y)\log p(y\mid x) = \sum_x p(x) H(Y\mid X=x).$$

Theorem: chain rule

$H(X,Y) = H(X) + H(Y\mid X)$.

Proof. Take $-\mathbb{E}$ of $\log p(X,Y) = \log p(X) + \log p(Y\mid X)$. ■

2 Kullback–Leibler Divergence

Definition: KL divergence

$$D_{\mathrm{KL}}(p\,\|\,q) = \sum_x p(x)\log\frac{p(x)}{q(x)} = \mathbb{E}_p\Big[\log\frac{p(X)}{q(X)}\Big].$$

Terms with $p(x) = 0$ contribute 0; if $p(x) > 0 = q(x)$ for some x the divergence is $+\infty$.

Theorem: Gibbs' inequality

$D_{\mathrm{KL}}(p\,\|\,q) \ge 0$, with equality iff $p = q$.

Proof. Sum over $S = \{x : p(x) > 0\}$. By Jensen for the concave log,

$$-D_{\mathrm{KL}}(p \,\|\, q) = \sum_{x \in S} p(x) \log \frac{q(x)}{p(x)} \le \log \sum_{x \in S} p(x) \frac{q(x)}{p(x)} = \log \sum_{x \in S} q(x) \le \log 1 = 0.$$

Equality forces (i) $q(x)/p(x) = c$ constant on S (Jensen) and (ii) $\sum_S q = 1$; then $c = \sum_S q = 1$, so $q = p$. Conversely $p = q$ gives $\log 1 = 0$ term by term. ■

Example: KL is not symmetric

For Bernoulli parameters, $D(r \,\|\, s) = r \log \frac{r}{s} + (1 - r) \log \frac{1-r}{1-s}$. With $r = 0.9$, $s = 0.5$: $D(r\|s) \approx 0.368$ but $D(s\|r) \approx 0.511$ nats. KL is not a metric (no symmetry, no triangle inequality).

3 Mutual Information

Definition: mutual information

$$I(X;Y) = D_{\mathrm{KL}}(p_{XY} \,\|\, p_X p_Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}.$$

Theorem: identities

$$I(X;Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X) = H(X) + H(Y) - H(X,Y) \ \ge 0,$$

with $I = 0$ iff $X \perp Y$. Consequently $H(X \mid Y) \le H(X)$ (conditioning reduces entropy).

Proof. $\log \frac{p(x,y)}{p(x)p(y)} = \log p(x \mid y) - \log p(x)$; take $\mathbb{E}$ and use $\sum_y p(x,y) = p(x)$ to get $-H(X \mid Y) + H(X)$. The last form follows from the chain rule. Nonnegativity and the equality case are Gibbs' inequality. ■

The lecture's entropy diagram encodes $H(X,Y) = H(X \mid Y) + I(X;Y) + H(Y \mid X)$. For a fair bit X and $Y = X$, $I = 1$ bit; for independent fair bits, $I = 0$.

4 Cross-Entropy

Definition: cross-entropy

$H(p,q) = -\sum_x p(x) \log q(x) = \mathbb{E}_p[-\log q(X)]$.

Theorem: cross-entropy = entropy + KL

$H(p,q) = H(p) + D_{\mathrm{KL}}(p \,\|\, q)$. Hence for fixed p, $\arg\min_q H(p,q) = \arg\min_q D_{\mathrm{KL}}(p \,\|\, q) = p$.

Proof. $-\sum p \log q = -\sum p \log p + \sum p \log(p/q)$. The second statement uses Gibbs. ■

Remark: classification loss and soft labels

For a one-hot label $\mathbf{e}_y$, $H(\mathbf{e}_y) = 0$ and $H(\mathbf{e}_y, q) = D_{\mathrm{KL}}(\mathbf{e}_y \| q) = -\log q_y$: the classification loss. $H(p,q)$ is *linear in its first argument*, $H(\alpha p + \beta p', q) = \alpha H(p,q) + \beta H(p',q)$, because

p enters only as coefficients. Therefore a soft label that averages n one-hot labels gives $H(\frac{1}{n}\sum_j \mathbf{e}_{y_j}, q) = \frac{1}{n}\sum_j(-\log q_{y_j})$: fitting the label distribution by KL is equivalent (up to the constant $H(\hat{p})$) to ordinary cross-entropy on every individual label (2024 midterm Q2).

5 Jensen–Shannon Divergence (PS1 Problem 1)

Definition: JS divergence

With $m = \frac{1}{2}(p+q)$, $D_{\mathrm{JS}}(p \,\|\, q) = \frac{1}{2}D_{\mathrm{KL}}(p \,\|\, m) + \frac{1}{2}D_{\mathrm{KL}}(q \,\|\, m)$.

Theorem: properties

(i) $D_{\mathrm{JS}}(p\|q) \geq 0$, with equality iff $p = q$.

(ii) $D_{\mathrm{JS}}(p\|q) = D_{\mathrm{JS}}(q\|p)$.

(iii) $D_{\mathrm{JS}}(p\|q) = H(m) - \frac{1}{2}H(p) - \frac{1}{2}H(q)$.

(iv) $D_{\mathrm{JS}}(p\|q) \leq \log 2$, with equality iff p and q have disjoint supports.

Proof. (i) Both KL terms are nonnegative by Gibbs, so $D_{\mathrm{JS}} \geq 0$. If $D_{\mathrm{JS}} = 0$ then $D_{\mathrm{KL}}(p\|m) = 0$, so $p = m = \frac{1}{2}(p+q)$, i.e. $p = q$; conversely $p = q$ gives $m = p$ and both terms vanish. Note that $D_{\mathrm{KL}}(p\|m) < \infty$ always, since $m \geq p/2$. (ii) m is symmetric in p, q and the definition is a symmetric sum. (iii) $D_{\mathrm{KL}}(p\|m) = -H(p) - \sum p \log m$; averaging with the q term gives $-\frac{1}{2}H(p) - \frac{1}{2}H(q) - \sum m \log m$. (iv) Substituting m,

$$D_{\mathrm{JS}} = \log 2 + \tfrac{1}{2}\sum p \log \frac{p}{p+q} + \tfrac{1}{2}\sum q \log \frac{q}{p+q} \leq \log 2,$$

since $p/(p+q) \leq 1$ and $q/(p+q) \leq 1$. Equality iff $q = 0$ wherever $p > 0$ and vice versa. ■

Remark: JS as mutual information

Let B be a fair bit and $Z \sim p$ if $B = 0$, $Z \sim q$ if $B = 1$. Then $Z \sim m$, $H(Z \mid B) = \frac{1}{2}H(p) + \frac{1}{2}H(q)$, and by (iii) $I(Z; B) = D_{\mathrm{JS}}(p\|q) \leq H(B) = \log 2$.